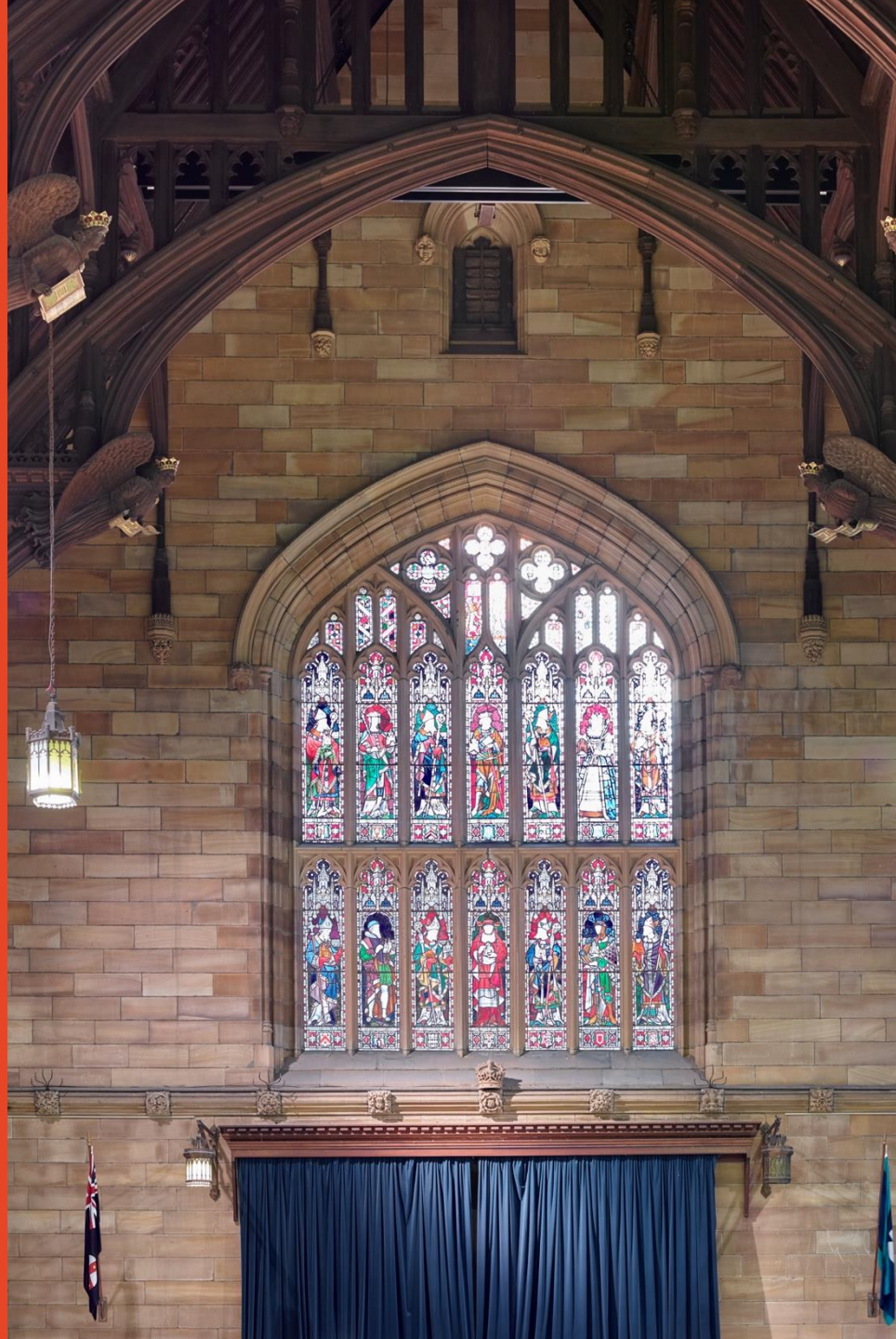
Data structures and Algorithms

Trees

Version 1.0

Dr Armin Chitizadeh

THE UNIVERSITY OF SYDNEY

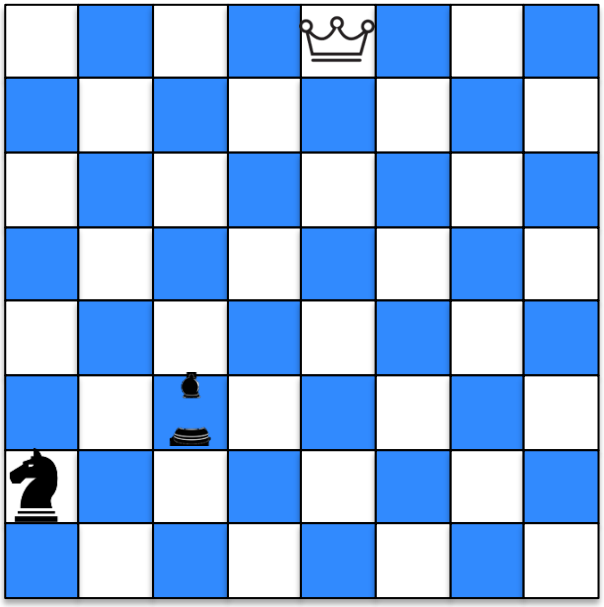


Respecting others

What is your favourite movie that has parallel worlds?

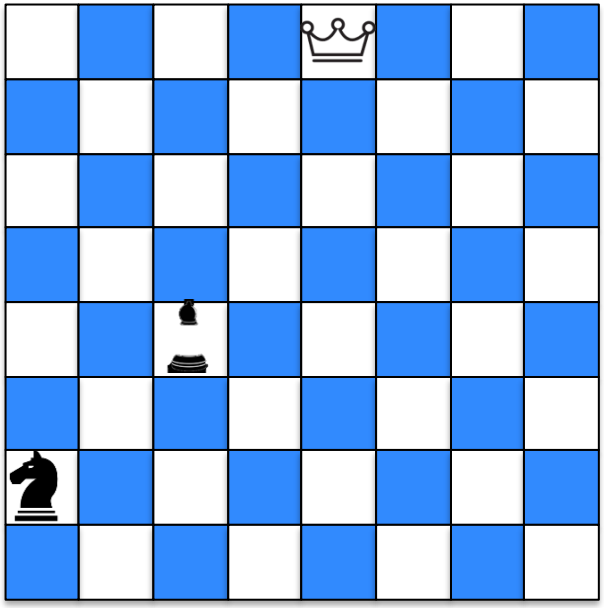
https://www.menti.com/al2a7js9zrpc

How to find optimal game strategy

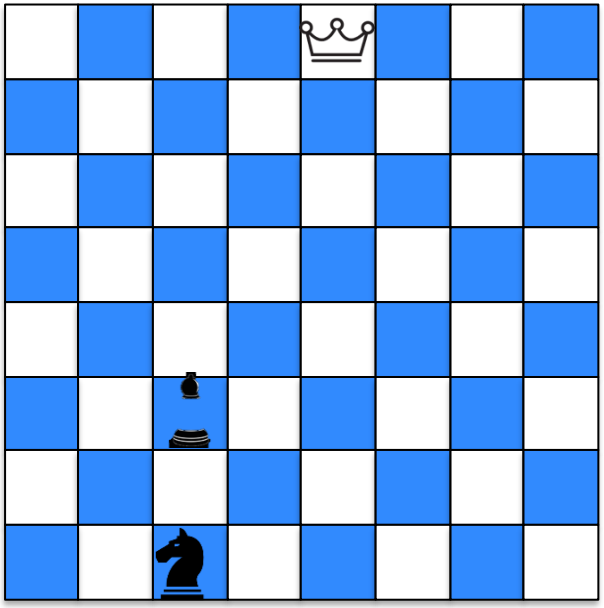


How to find optimal game strategy

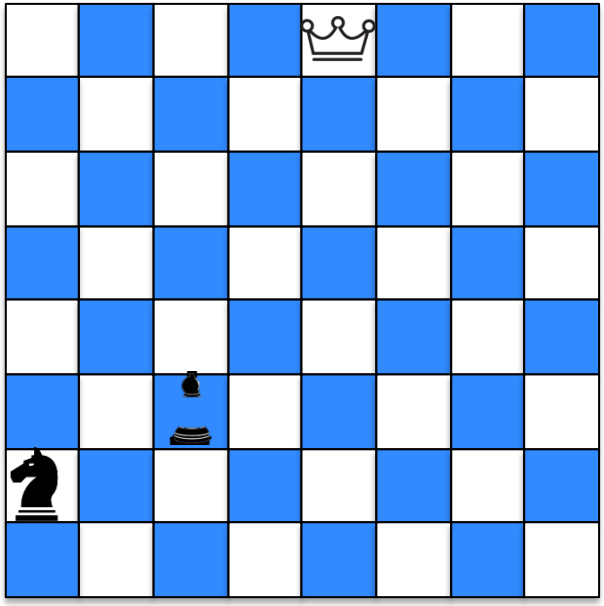
AA



AB

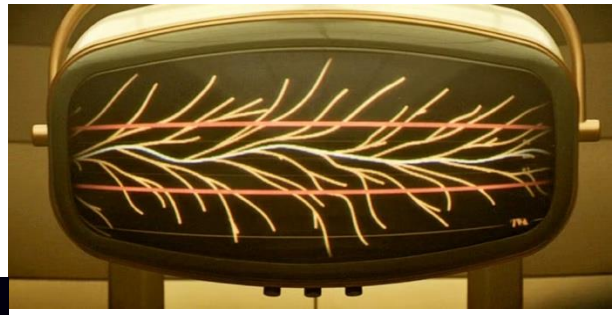


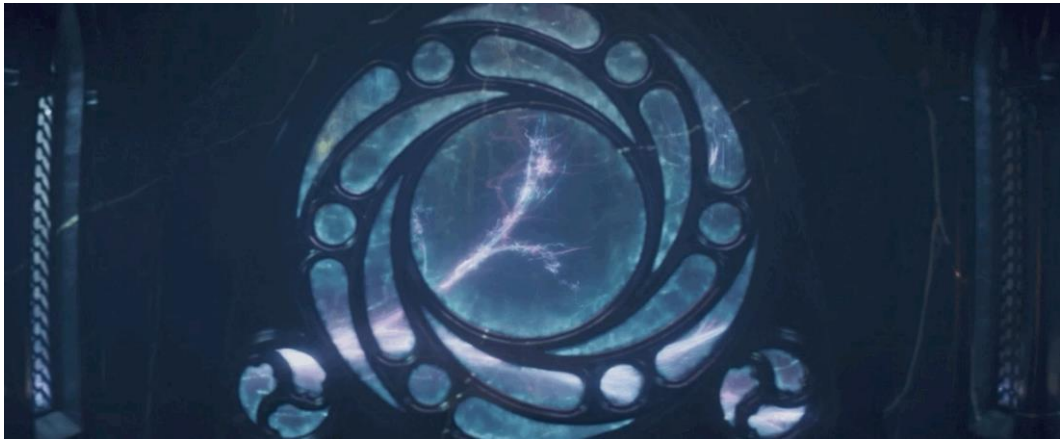
A

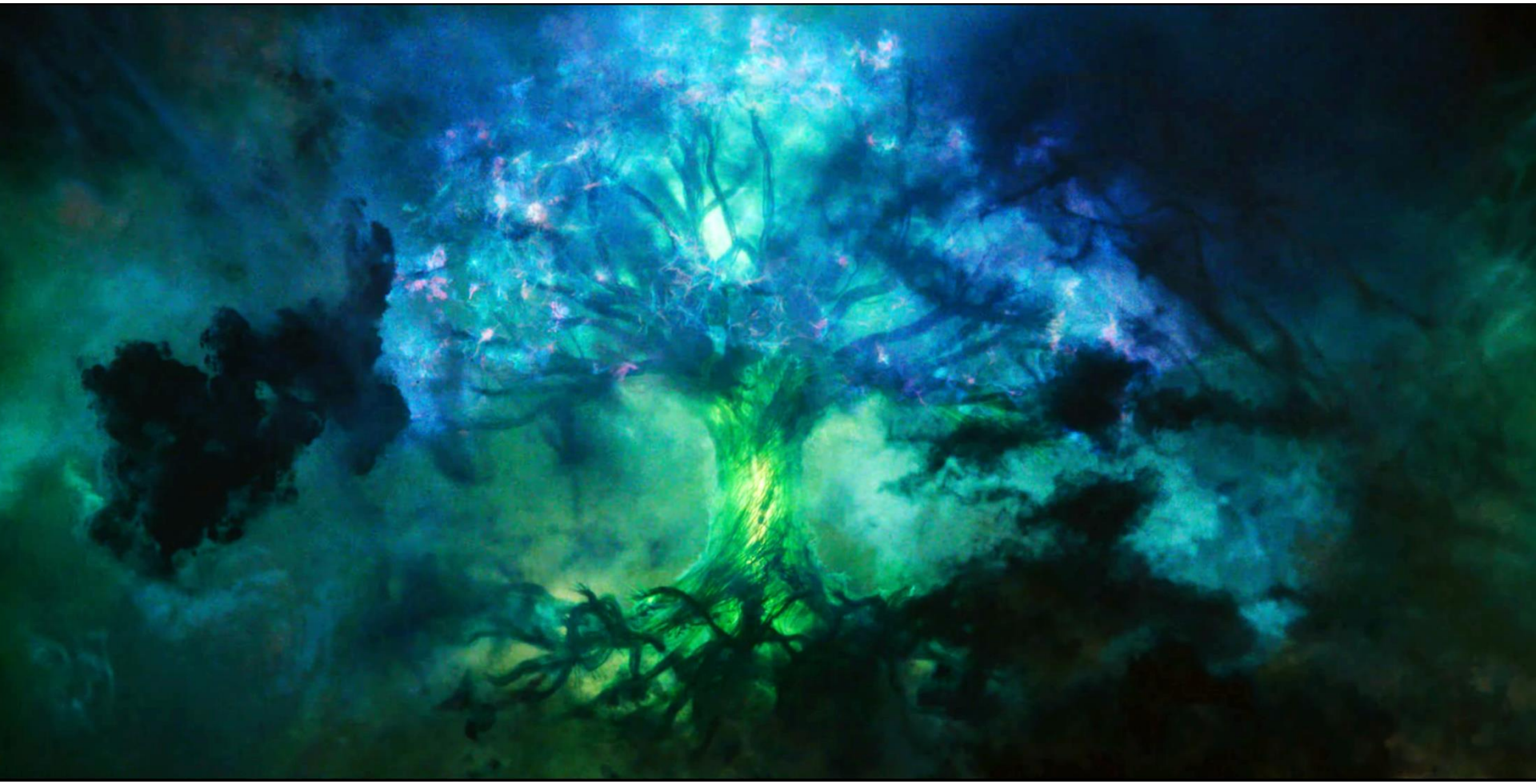


Can we fit it into a list?

What should we do?

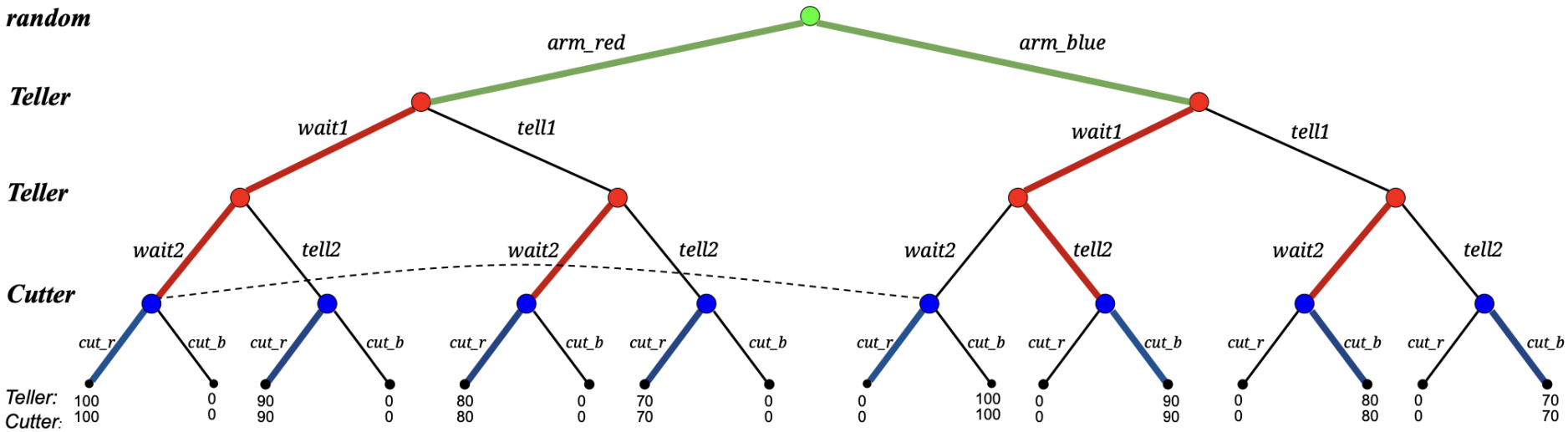






Tree

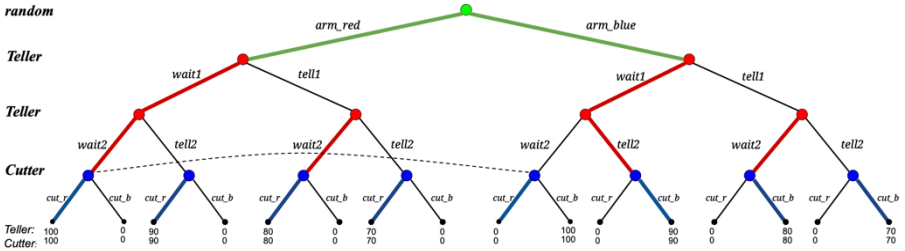
The Game Tree of Partially Hidden Extending Cutting Wire Game



(b) The Nash equilibrium of waiting when the red wire arms the bomb.

Agenda: Trees

- Definition and terminology
- Applications
- Tree ADT
- Tree traversal algorithms
- Binary trees
- Implementing trees
- Recursive code on trees

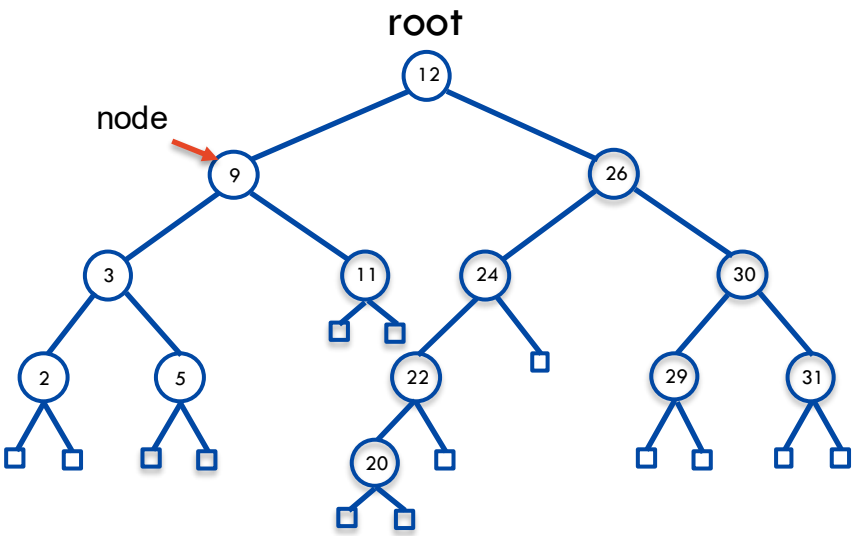


(b) The Nash equilibrium of waiting when the red wire arms the bomb.

Trees

leaves

root



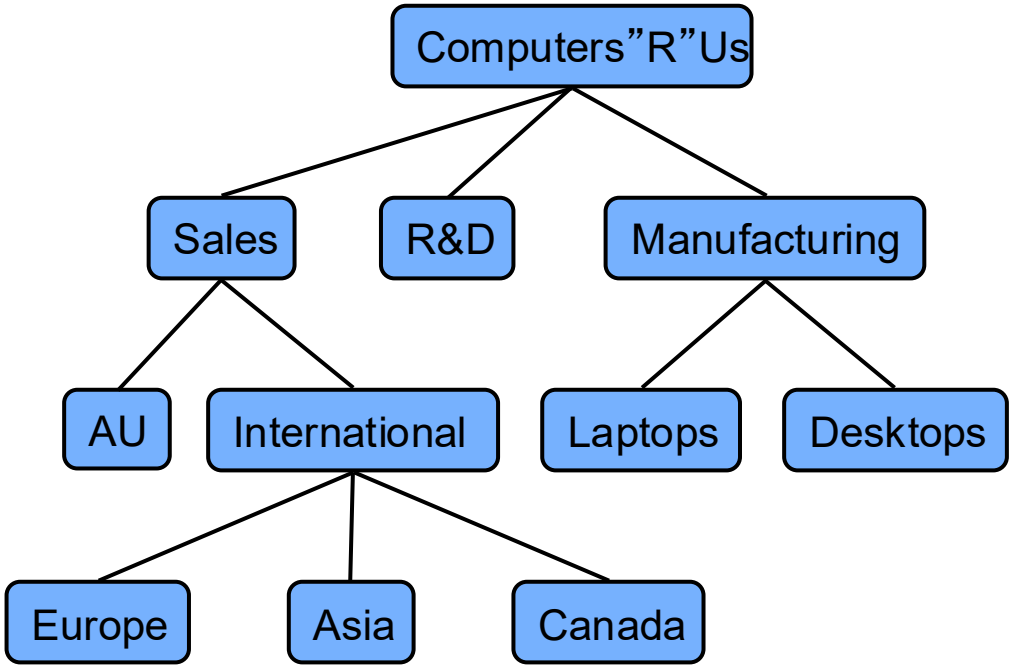
leaves

What is a Tree

In computer science, a tree is an abstract model of a <u>hierarchical</u> structure

A tree consists of nodes with a parent-child relation

- if u is parent of v, then v is a child of u
- a node has at most **one** parent in a tree
- a node can have zero, one or more children



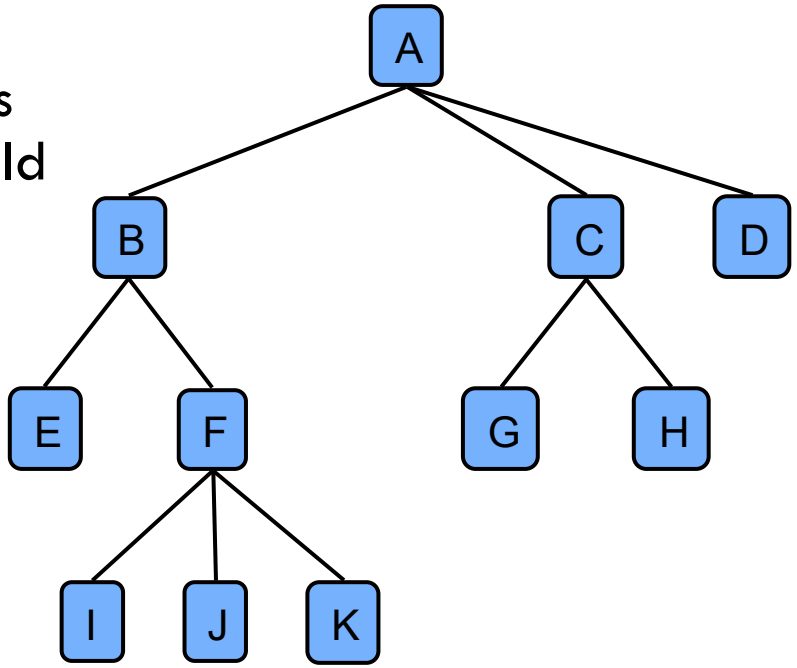
Applications:

- Organization charts
- File systems
- Phrase structure

Formal definition

A tree T is made up of a set of nodes endowed with parent-child relationship with following properties:

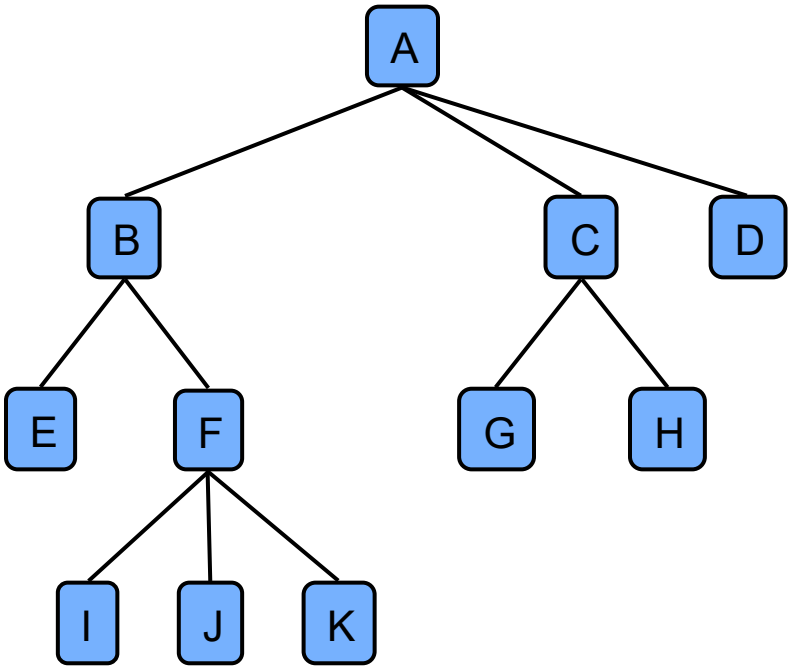
- If T is non-empty, it has a special node called the root that has no parent
- Every node v of T other than the root has a unique parent
- Following the parent relation always leads to the root (i.e., the parent-child relation does not have "cycles")



Tree Terminology

Depending on where they are in the tree, we classify nodes into:

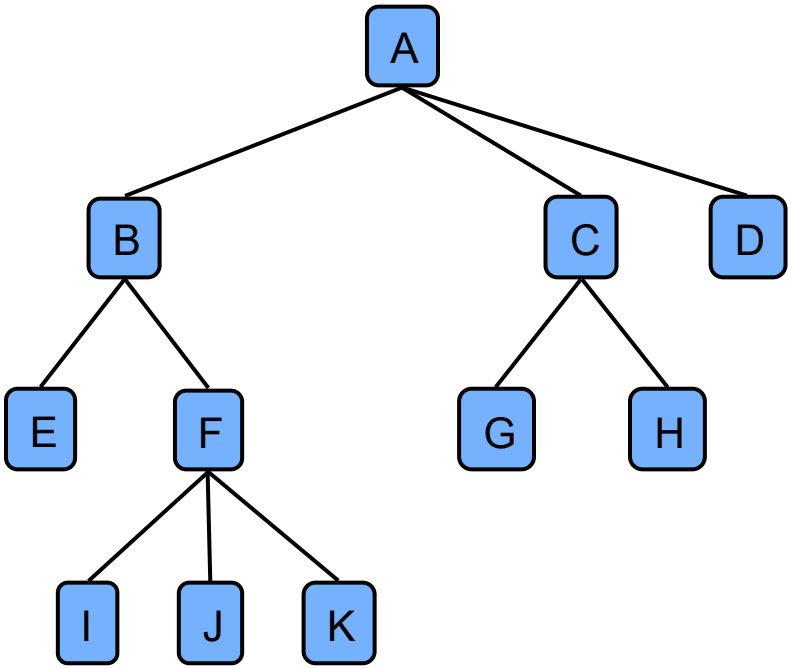
- Root: node without parent (e.g., A)
- Internal node: node with at least one child (e.g., A, B, C, F)
- External/leaf node: node without children (e.g., E, I, J, K, G, H, D)



Tree Terminology

We can extend the parent-child relation to capture indirect relations:

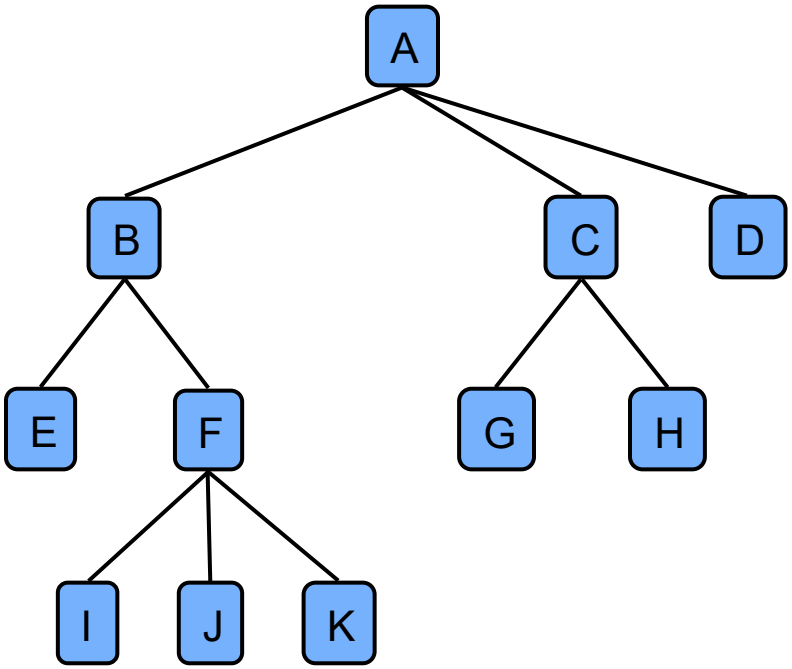
- Ancestors: parent, grandparent, great-grandparent, etc. (e.g., ancestors of F are A, B)
- Descendants: child, grandchild, great-grandchild, etc. (e.g., descendants of B are E, F, I, J, K)
- Two nodes with the same parent are siblings (e.g., B and D)



Tree Terminology

More fine-grained location concepts:

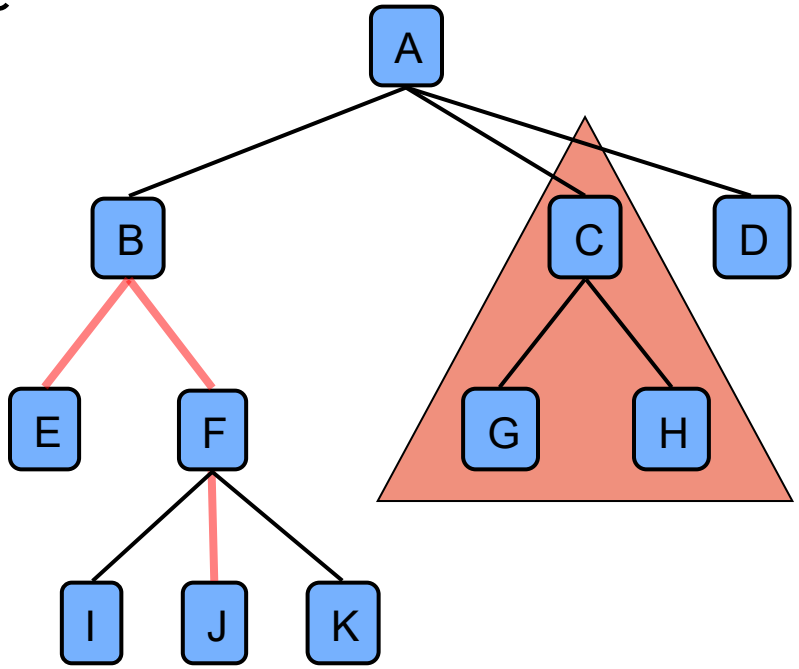
- Depth of a node: number of ancestors not including itself (e.g., depth(F) = 2)
- Level: set of nodes with given depth (e.g., {E, F, G, H} are level 2)
- Height of a tree: maximum depth (e.g., 3)



Tree Terminology

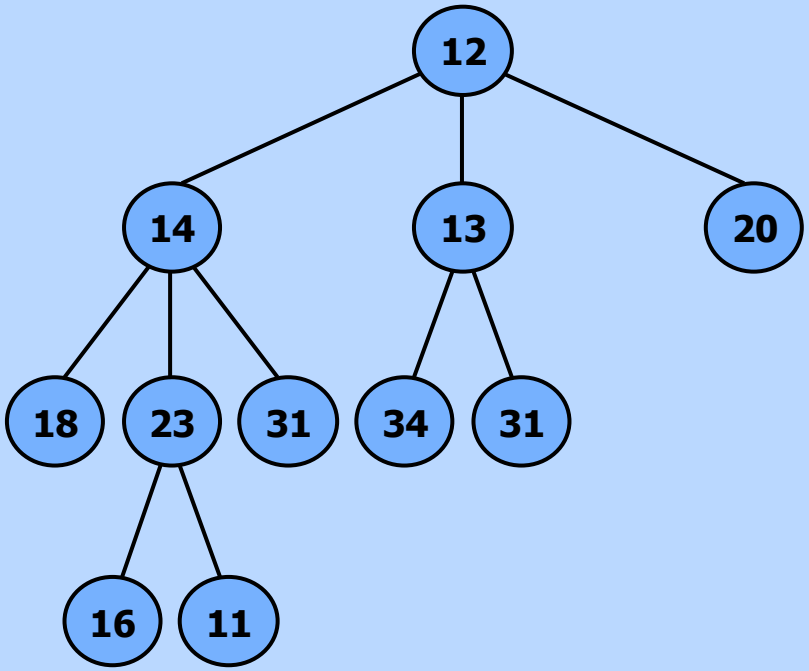
Substructures of a tree:

- Subtree: tree made up of some node and its descendants. (e.g., subtree rooted at C is {C, G, H})
- Edge: pair of nodes (u, v) such that one is the parent of the other
- Path: sequence of nodes such that 2 consecutive nodes in the sequence have an edge (e.g., <E, B, F, J>).



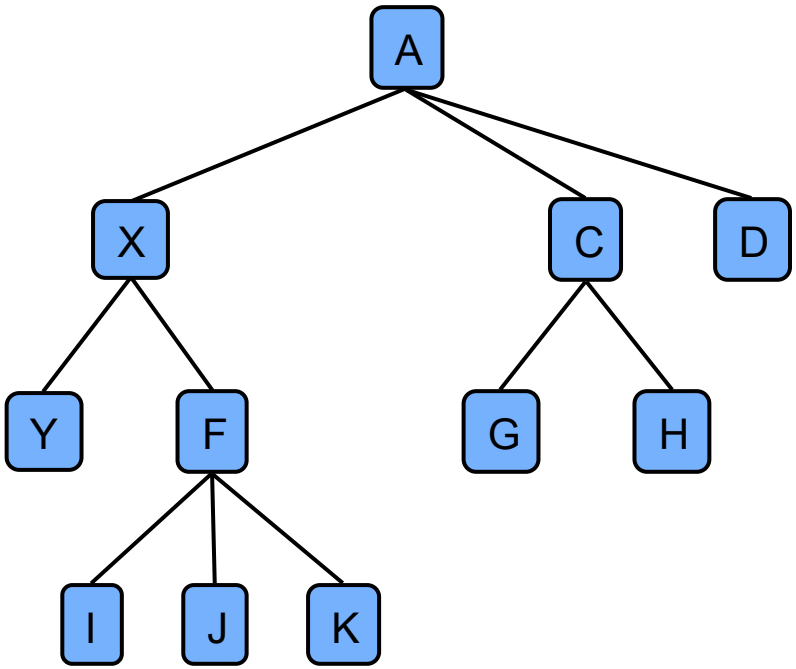
Interactive Example

- Node 14 has depth … **1**
- The tree has height … **3**
- Subtree rooted at node 14 has height … **2**
- Any subtree of a leaf has height … **0**
- The root has depth … **0**



Tree facts

- If node X is an ancestor of node Y, then Y is a descendant of X.
- Ancestor/descendant relations are transitive
- Every node is a descendant of the root
- There may be nodes where neither is an ancestor of the other
- Every pair of nodes has at least one common ancestor.
- The lowest common ancestor (LCA) of x and y is a node z such that z is the ancestor of x and y and no descendant of z has that property



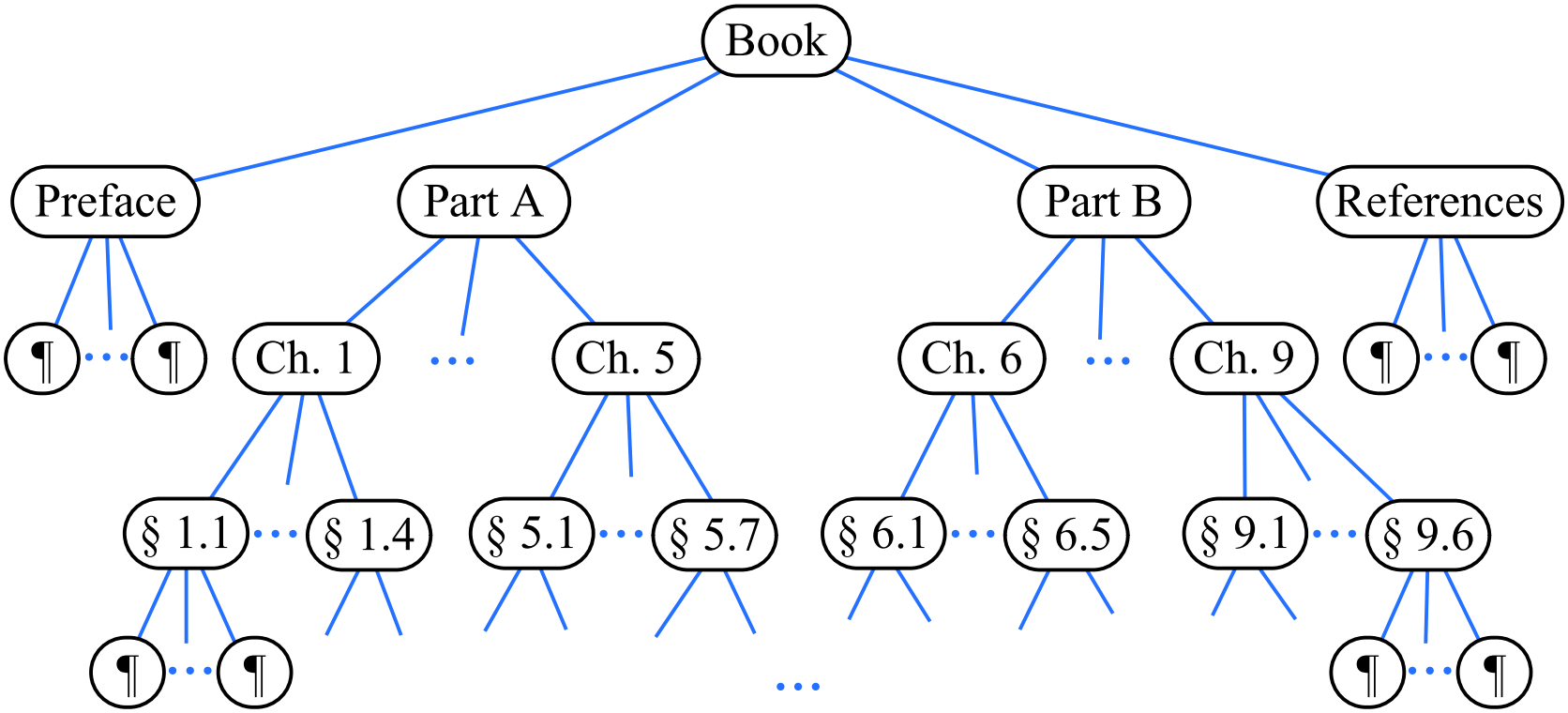
Ordered Trees

Sometimes order of siblings matter

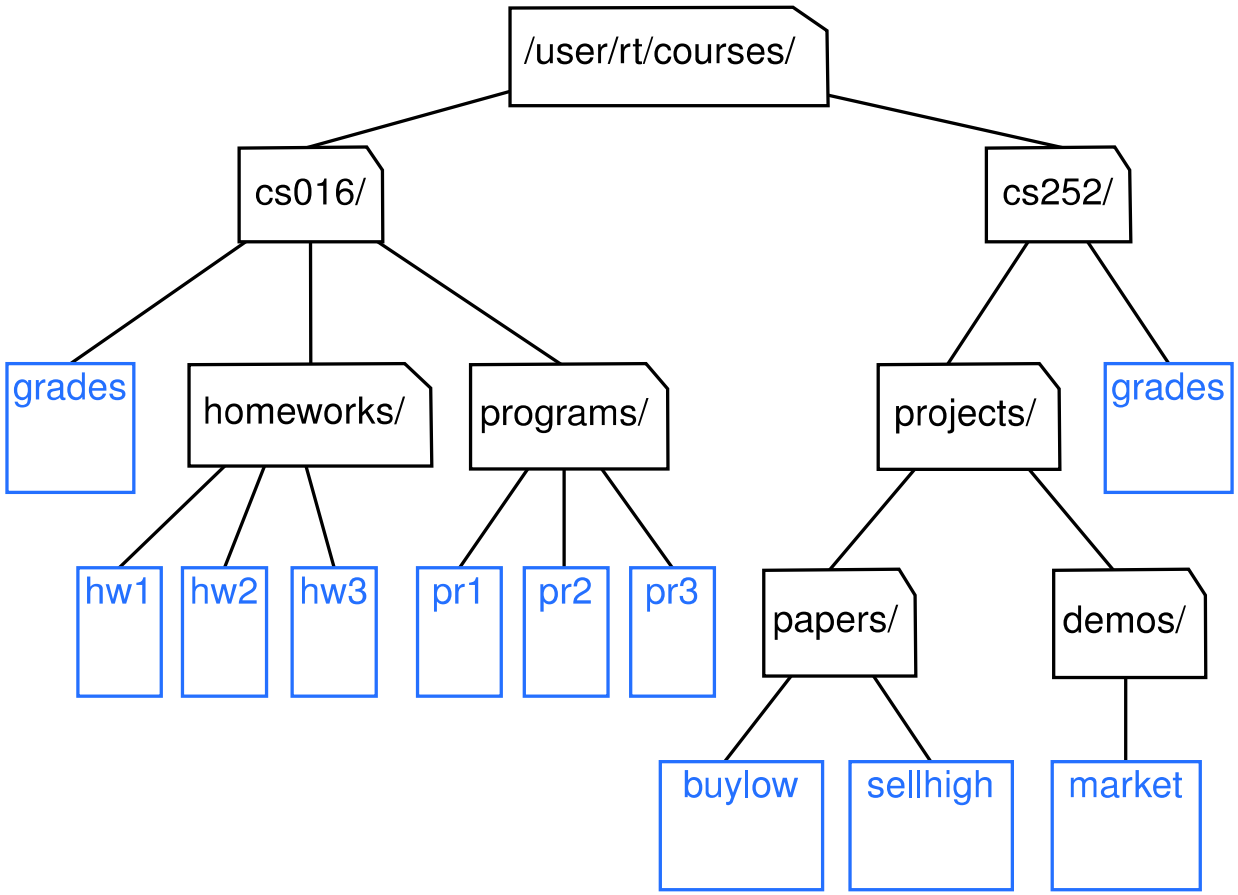
In an ordered tree there is a prescribed order for each node's children

In a diagram this ordering is usually represented by the left to right arrangement of the nodes

Application: Document structure



Application: OS file structure



Challenge 4A

List as many examples of trees from daily life. Bonus if you list the items in the hidden list below!

Examples

Tree ADT

- Position as Node abstraction
- Generic methods:
  - integer size()
  - boolean isEmpty()
  - Iterator iterator()
  - Iterable positions()
- Access methods:
  - Position root()
  - Position parent(p)
  - Iterable children(p)
  - Integer numChildren(p)

- Query methods:
  - boolean isInternal(p)
  - boolean isExternal(p)
  - boolean isRoot(p)

- Additional update methods may be defined by data structures implementing the Tree ADT

Node object

Node object implementation typically has the following attributes:

- value: the value associated with this Node
- children: set or list of children of this Node
- parent: (optional) the parent of this Node

```
def is_external(v)
    # test if v is a leaf
    return v.children.is_empty()

def is_root(v)
    # test if v is the root
    return v.parent = null
```

Traversing trees

A traversal visits the nodes of a tree in a systematic manner

When traversing a simpler structure like a list there is one natural traversal strategy (forward or backwards)

Trees are more complex and admit more than one natural way:

- pre-order
- post-order
- in-order (for binary trees)

Preorder Traversal

To do a preorder traversal starting at a given node, we visit the node <u>before</u> visiting its descendants

```
def pre_order(v)
    visit(v)
    for each child w of v
        pre_order(w)
```

If tree is ordered visit the child subtrees in the prescribed order

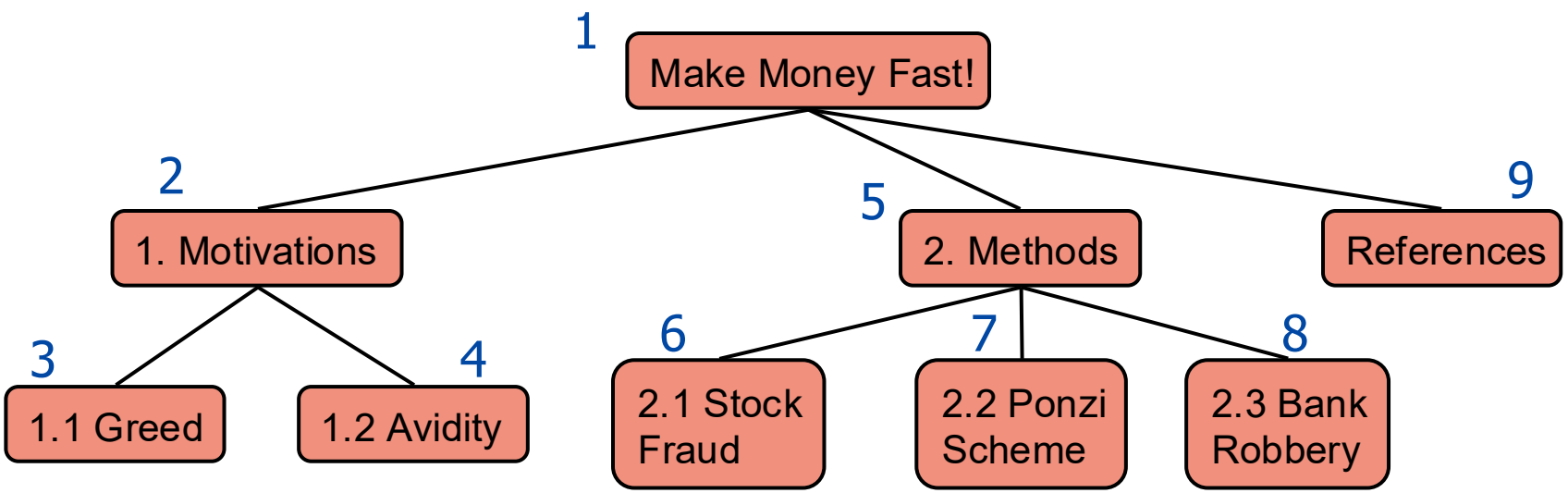
Visit does some work on the node:

- print node data
- aggregate node data
- modify node data

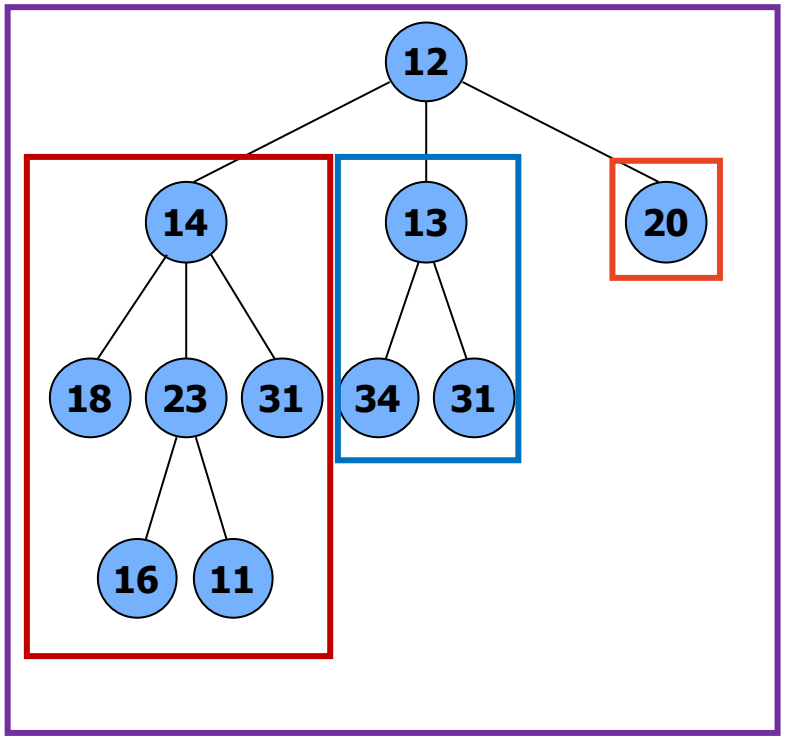
Preorder Traversal Example

Nodes are numbered in the order they are visited when we call pre_order() at the root

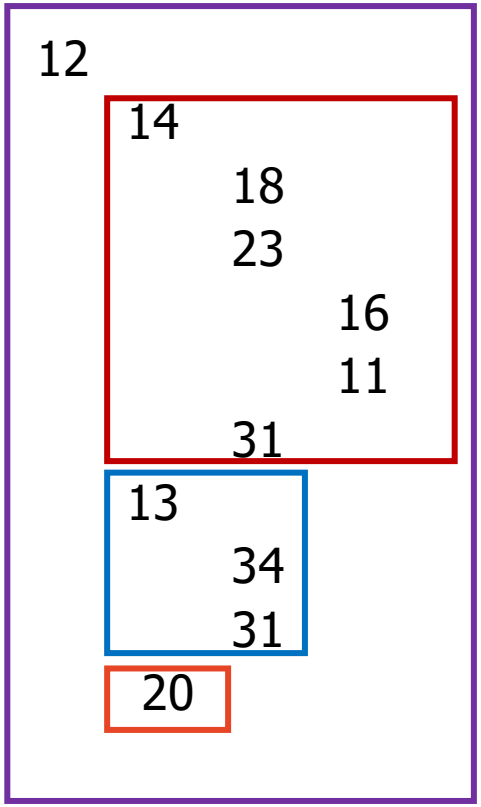
```
def pre_order(v)
    visit(v)
    for each child w of v
        pre_order(w)
```



Preorder Traversal Example



visit order



Preorder traversal of subtree

Postorder Traversal

To do a postorder traversal starting at a given node, we visit the node <u>after</u> its descendants

If tree is ordered visit the child subtrees in the prescribed order

Visit does some work on the node:

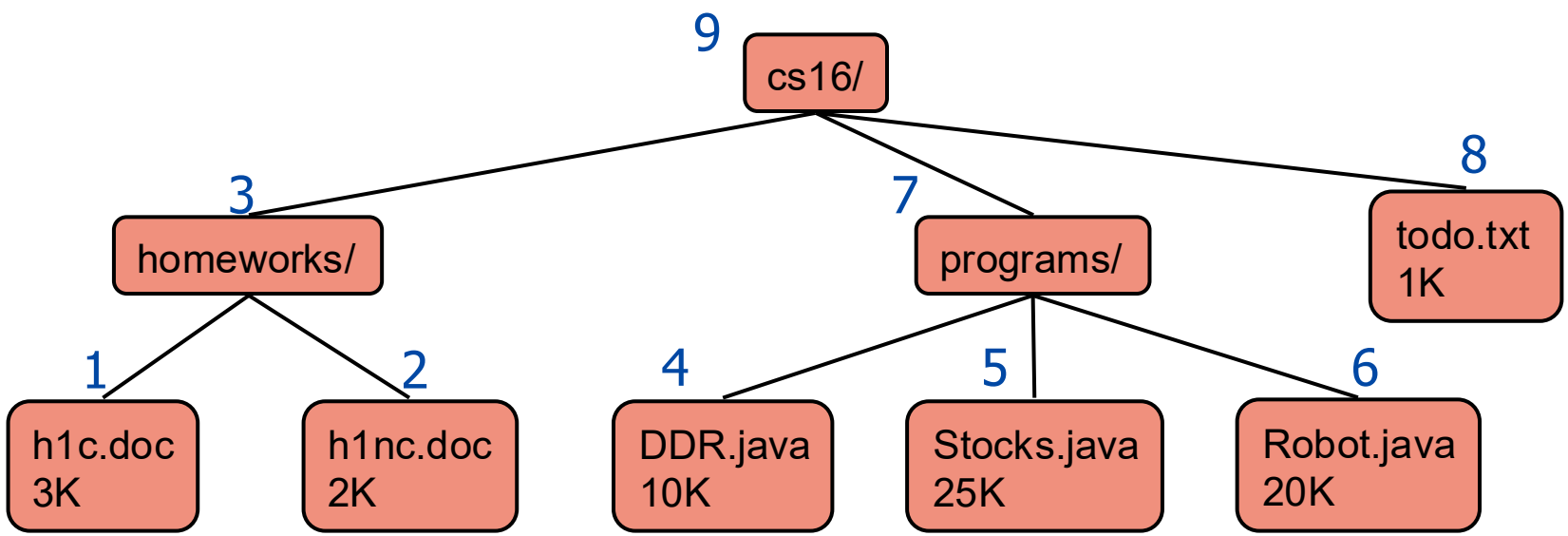
- print node data
- aggregate node data
- modify node data

```
def post_order(v)
    for each child w of v
        post_order(w)
    visit(v)
```

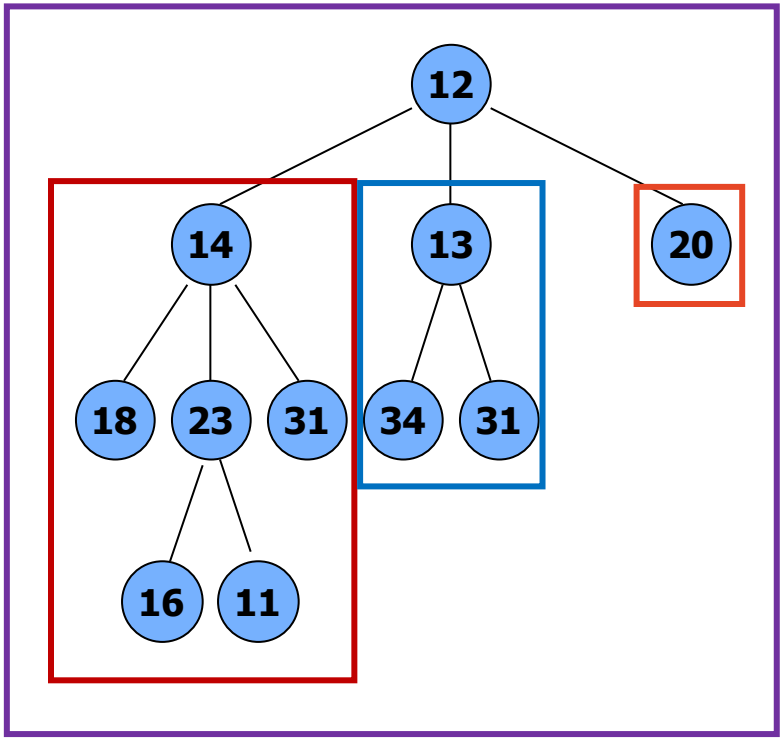
Postorder Traversal

Nodes are numbered in the order they are visited when we call post_order() at the root

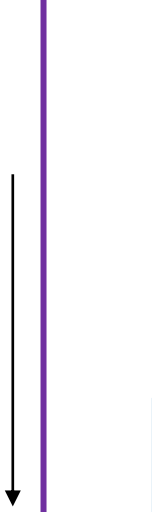
```
def post_order(v)
    for each child w of v
        post_order(w)
    visit(v)
```

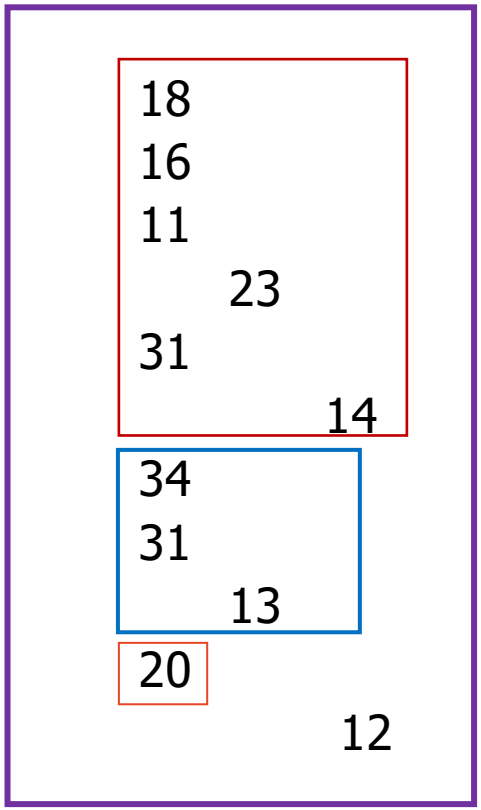


Traversing in postorder



visit order





Postorder traversal of subtree

Challenge 4B

What type of traversal would you use for the following examples?

1. File System Backup
2. Company Strategic Decisions
3. Chatbot Interaction
4. Cleaning a Room
5. Solving an arithmetic expression

?

Menti

- https://www.menti.com/al99iff6jn4v

Break!

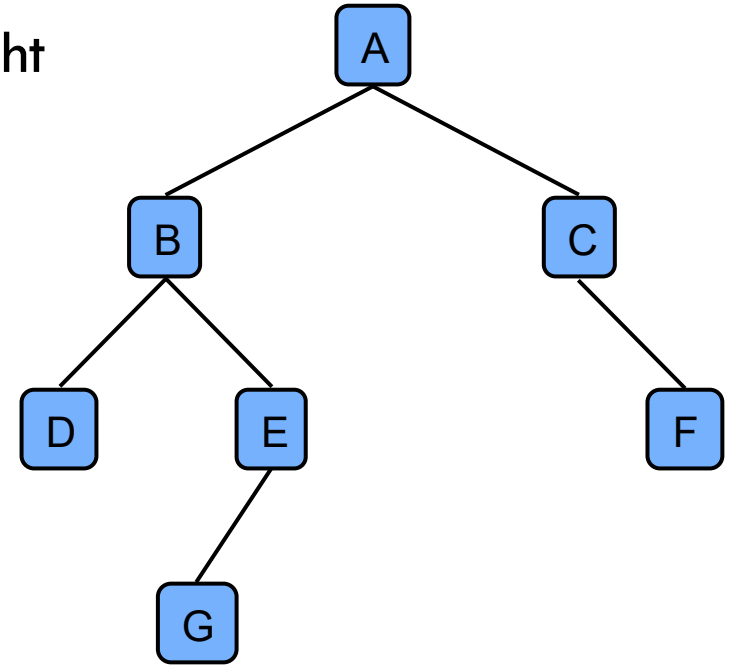
Binary Trees

A binary tree is an ordered tree with the following properties:

- Each internal node has at most two children
- Each child node is labeled as a left child or a right child
- Child ordering is left followed by right

The right/left subtree is the subtree root at the right/left child.

We say the tree is proper if every internal node has two children

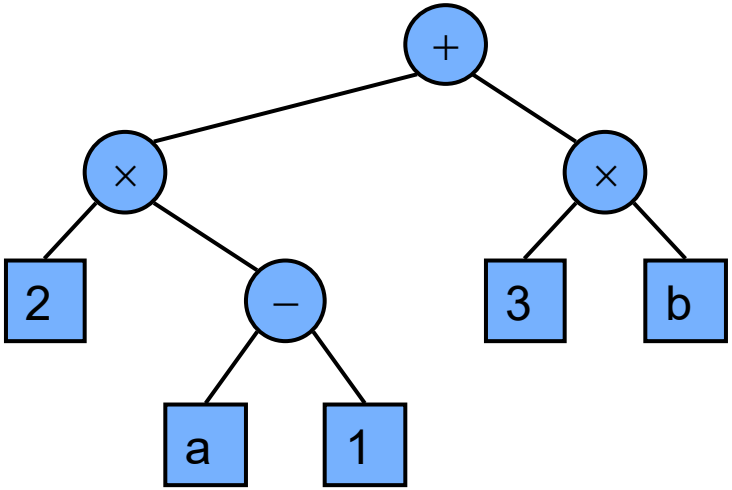


Binary tree application: Arithmetic expression tree

Binary tree associated with an arithmetic expression

- internal nodes: operators
- external nodes: operands

Example: Arithmetic expression tree for $(2 \times (a - 1) + (3 \times b))$

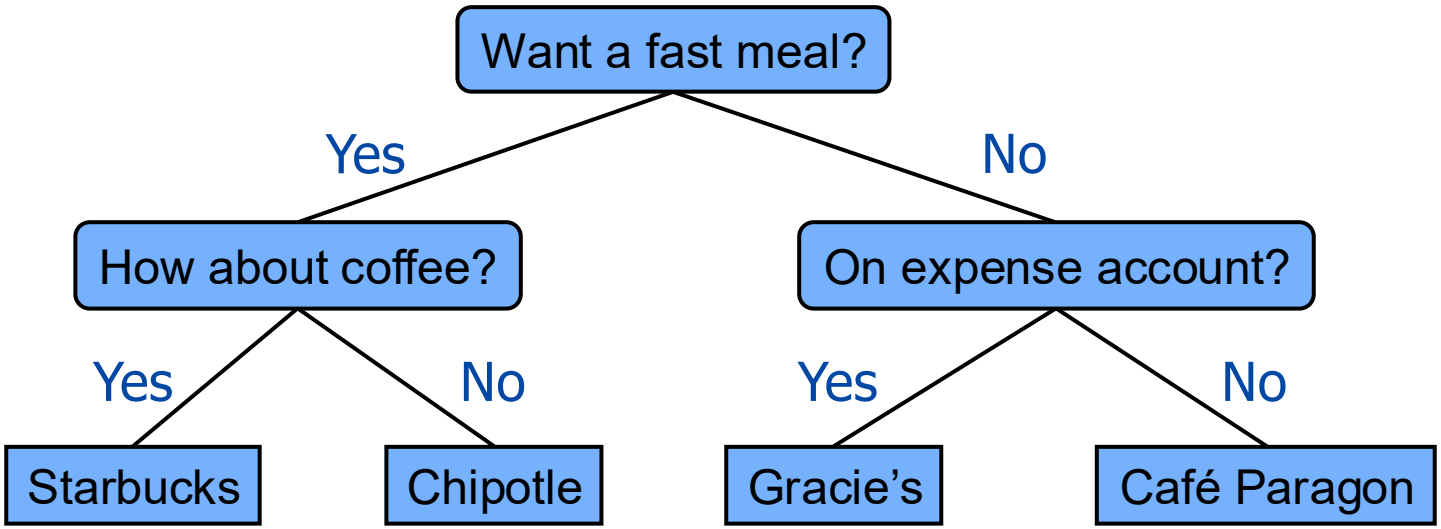


Binary tree application: Decision trees

Tree associated with a decision process

- internal nodes: questions with yes/no answer
- external nodes: decisions

Example: dining decision



Binary Tree Operations

- A binary tree extends the Tree operations, i.e., it inherits all the methods of a tree.
- Update methods may be defined by data structures implementing the binary tree
- Additional methods:
  - position leftChild(p)
  - position rightChild(p)
  - position sibling(p)

return null when there is no left, right, or sibling of p, respectively

Node object

Node object implementation typically has the following attributes:

- value: the value associated with this Node
- left: left child of this Node
- right: right child of this Node
- parent: (optional) the parent of this Node

```
def is_external(v)
    # test if v is a leaf
    return v.left = null and v.right = null
```

Challenge 4C

Which of these are binary trees?

1. Quiz Game with only True and False Answers
2. Corporate Company Structure
3. Elevator System
4. Password Validation

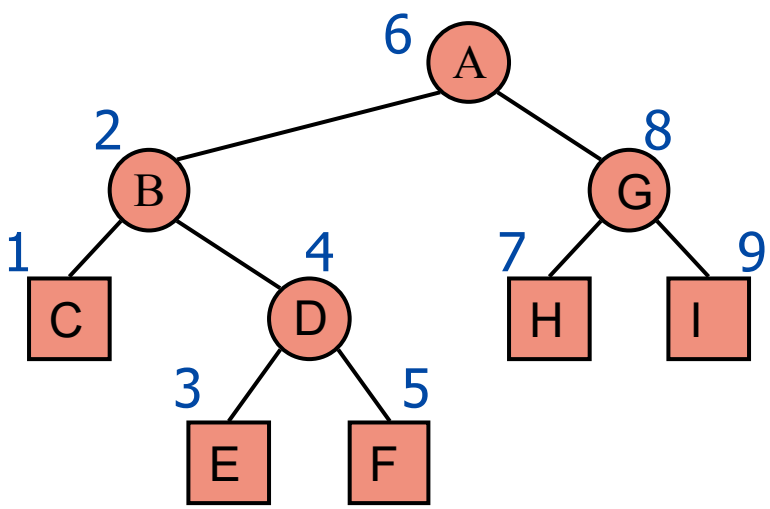
Inorder Traversal

To do an inorder traversal starting at a given node, the node is visited <u>after</u> its left subtree but <u>before</u> its right subtree

Visit does some work on the node:

- print node data
- aggregate node data
- modify node data

```
def in_order(v)
    if v.left ≠ null then
        in_order(v.left)
    visit(v)
    if v.right ≠ null then
        in_order(v.right)
```

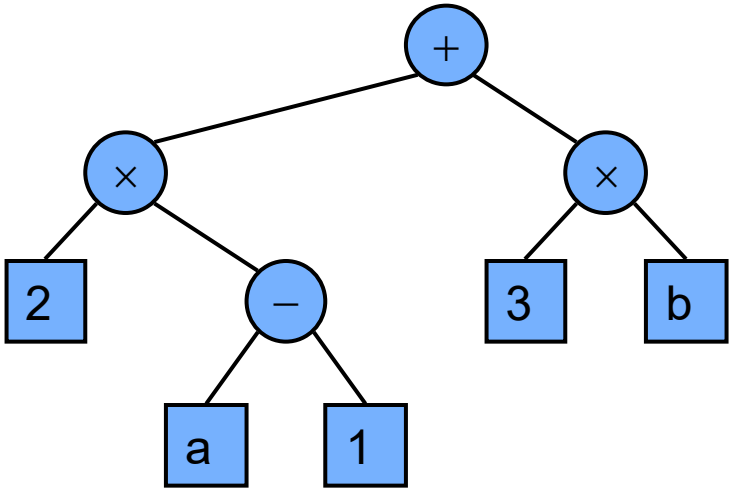


Print Arithmetic Expressions

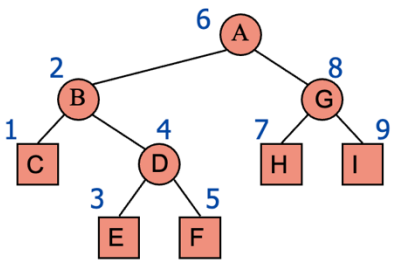
Extended inorder traversal:

- print operand or operator when visiting node
- print "(" before left subtree
- print ")" after right subtree

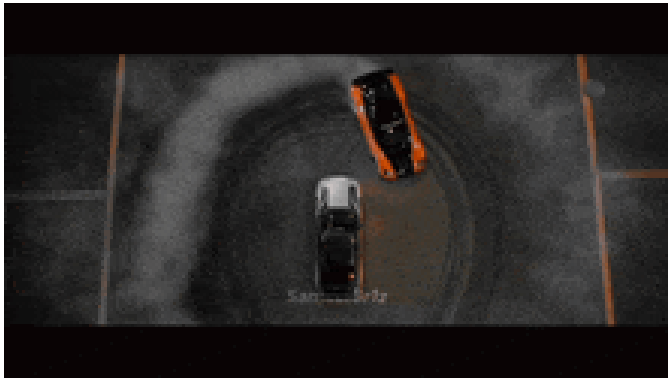
```
def print_expr(v)
    if v.left ≠ null then
        print("(")
        print_ expr(v.left)
    print(v.element)
    if v.right ≠ null then
        print_expr(v.right)
        print (")")
```



$$((2 \times (a - 1)) + (3 \times b))$$



Euler Tour Traversal

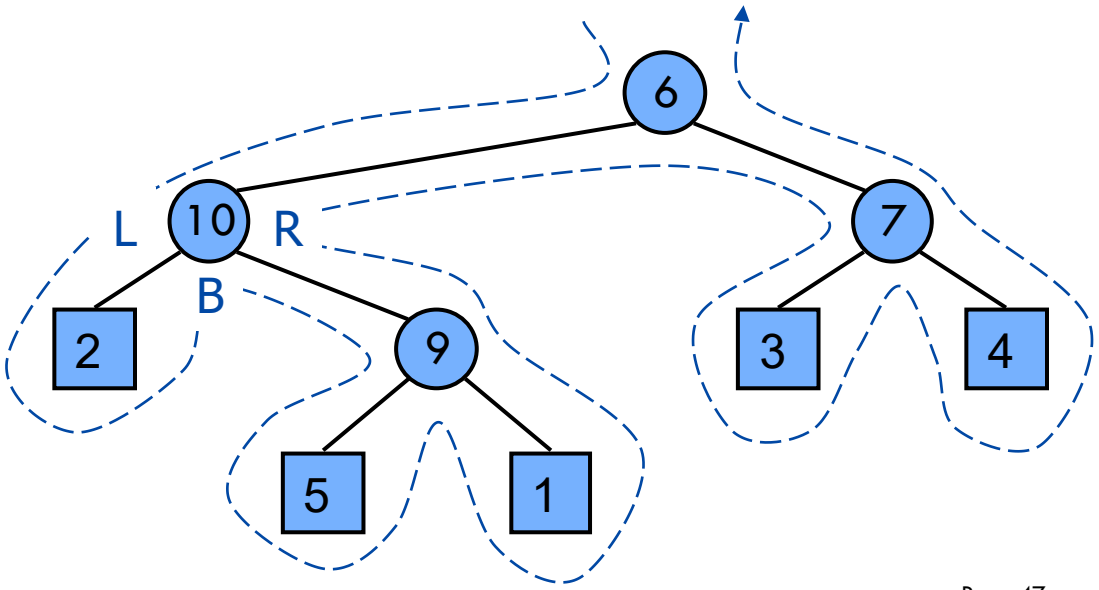


Euler Tour Traversal

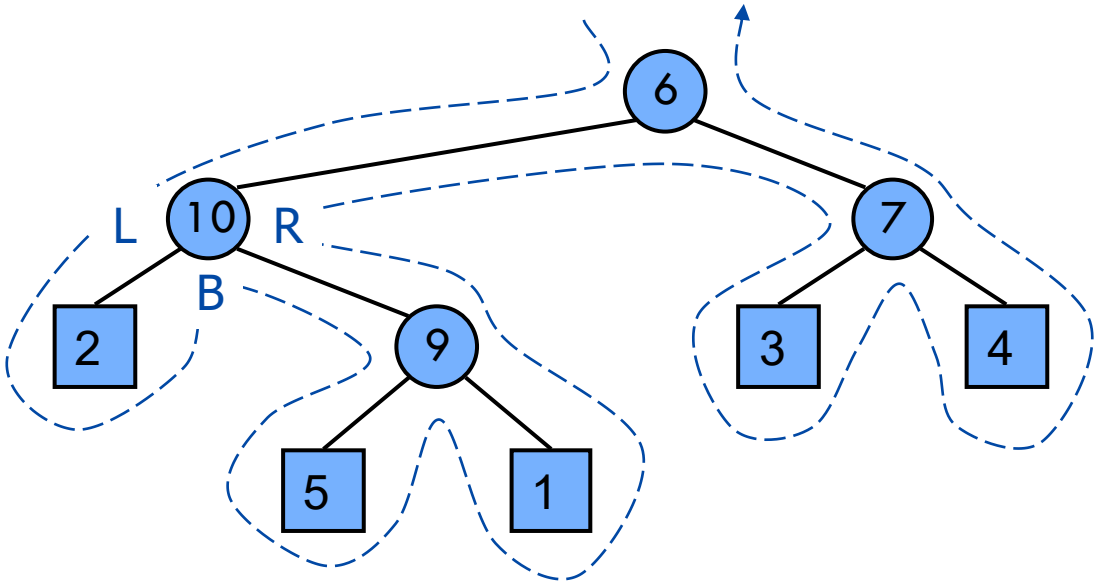
Generic traversal of a binary tree. Includes as special cases the preorder, postorder and inorder traversals

Walk around the tree, keeping the tree on your left, and visit each node three times:

- on the left (preorder)
- from below (inorder)
- on the right (postorder)



Euler Tour Traversal



6,10,2,2,2,10,9,5,5,5,9,1,1,1,9,10,6,7,3,3,3,7,4,4,4,7,6

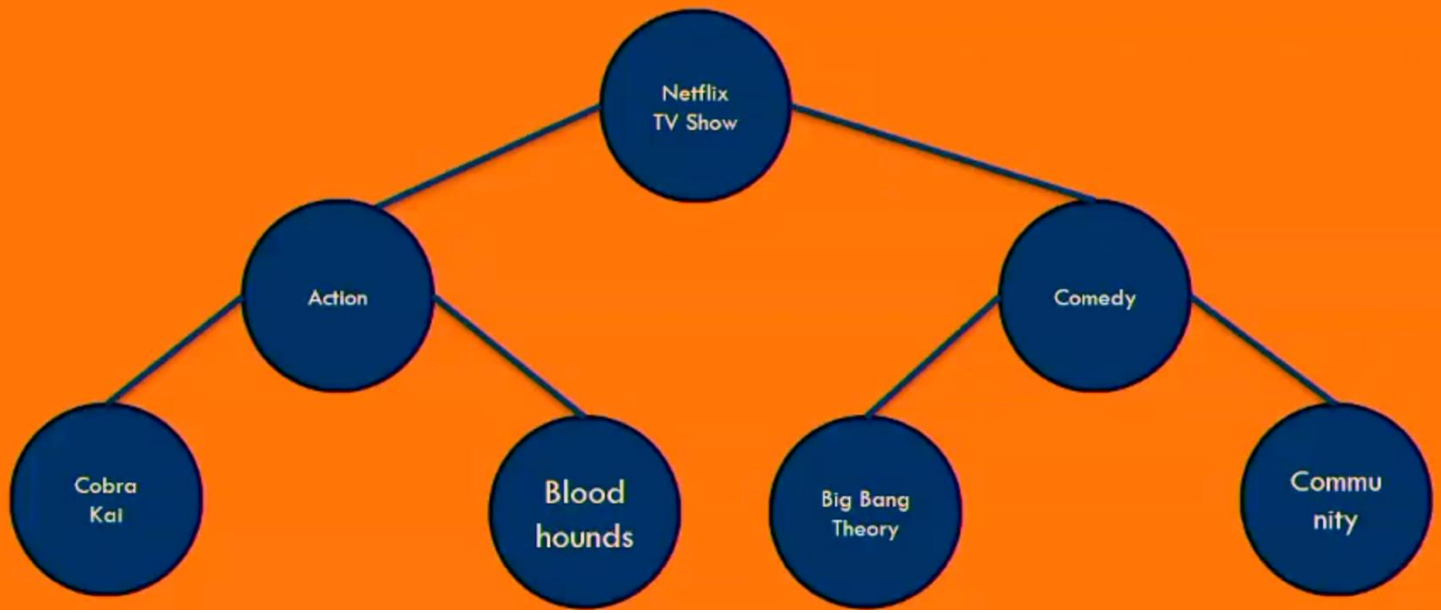
Preorder (first visit): 6, 10, 2, 9, 5, 1, 7, 3, 4

Inorder (second visit): 2, 10, 5, 9, 1, 6, 3, 7, 4

Postorder (third visit): 2, 5, 1, 9, 10, 3, 4, 7, 6

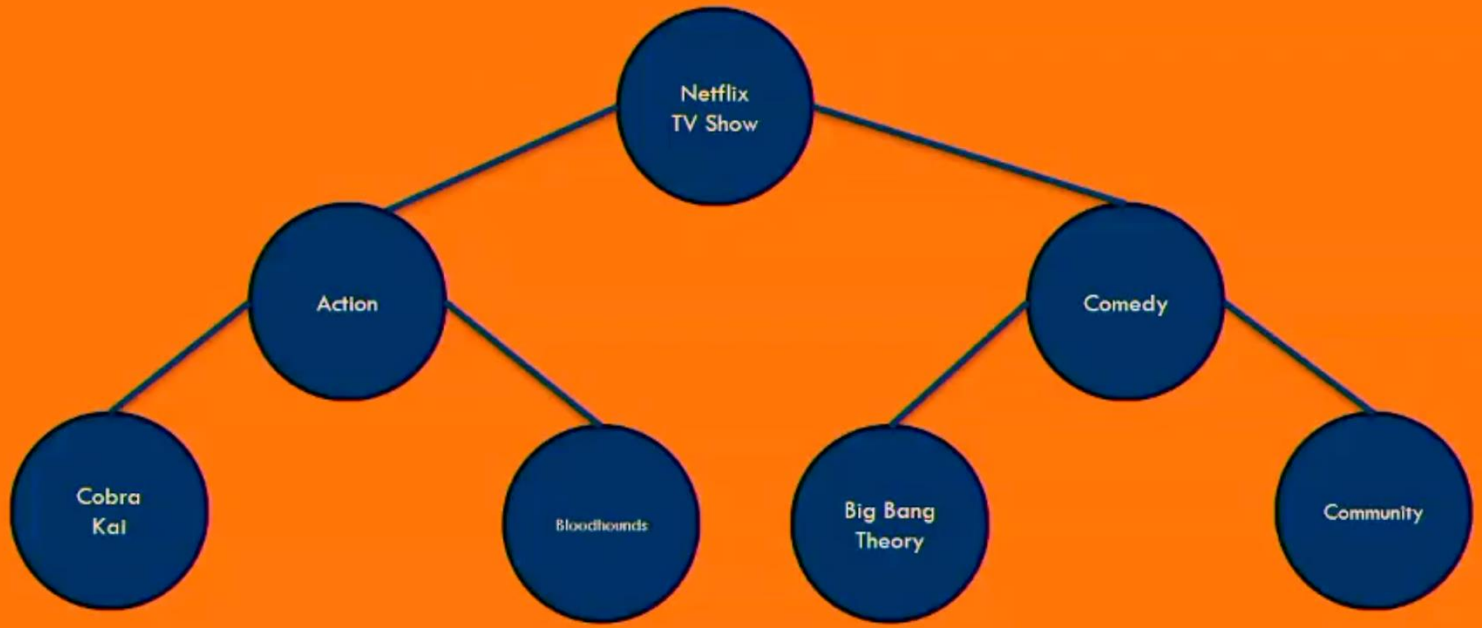
Challenge 4D

Write a list showing the **in-order traversal** of nodes for this tree (use Euler Tour if necessary).



Challenge 4D

[Cobra Kai, Action, Bloodhounds, Root, Big Bang Theory, Comedy, Community]

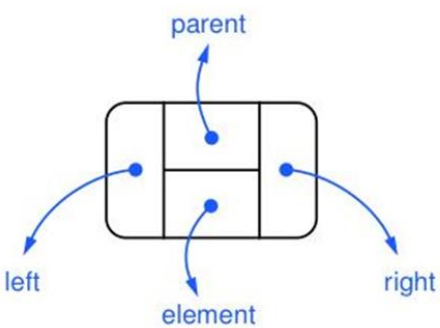


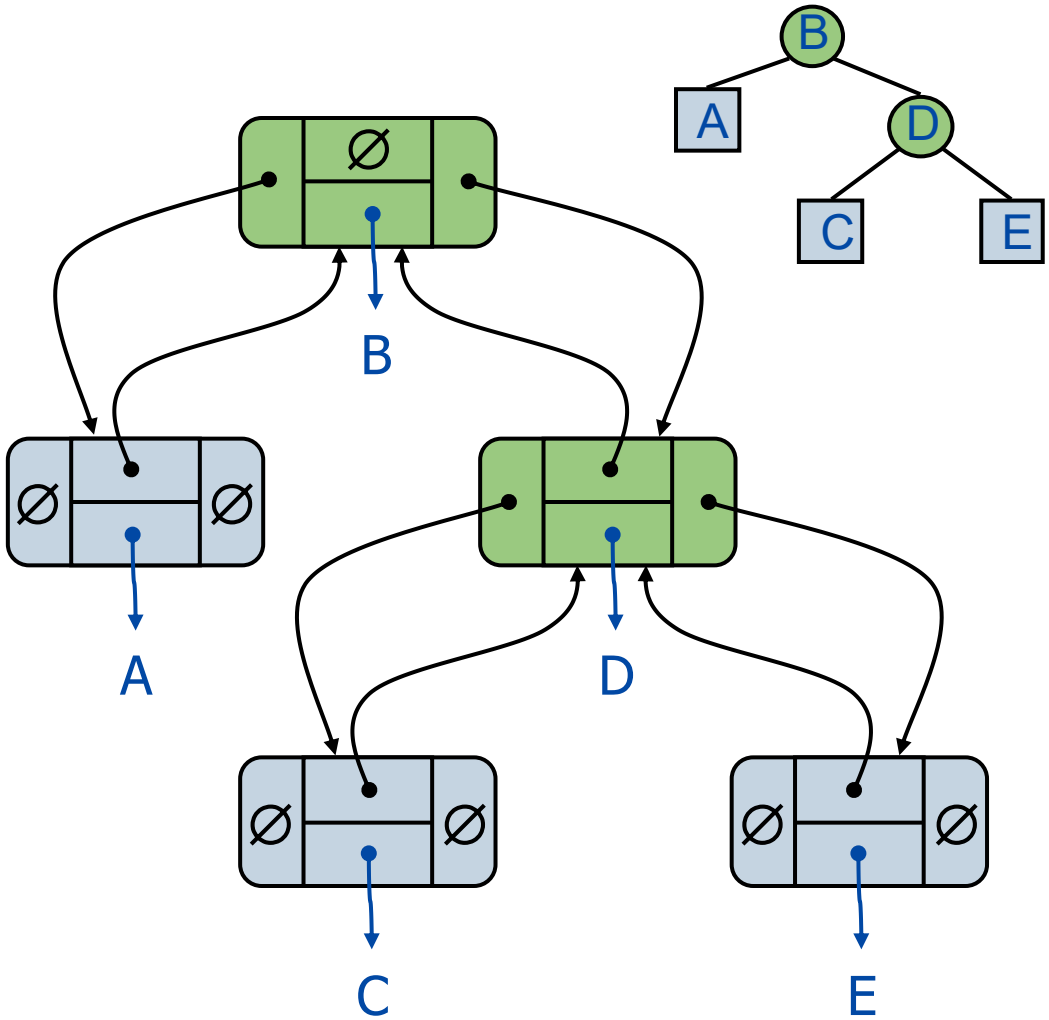
Linked Structure for Binary Trees

A node is represented by an object storing

- Element
- Parent node
- Left child node
- Right child node

Node objects implement the Position ADT





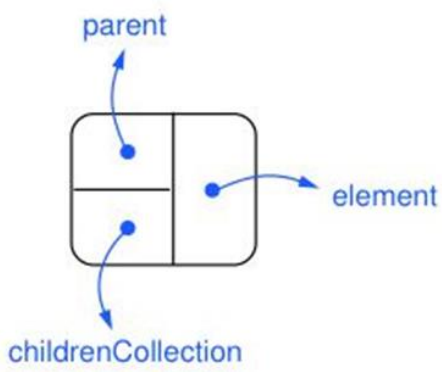
© Goodrich, Tamassia, Goldwasser

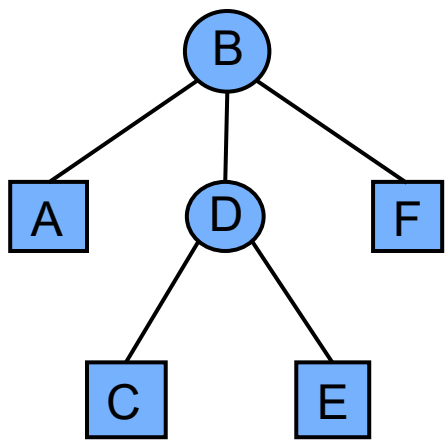
Linked Structure for General Trees

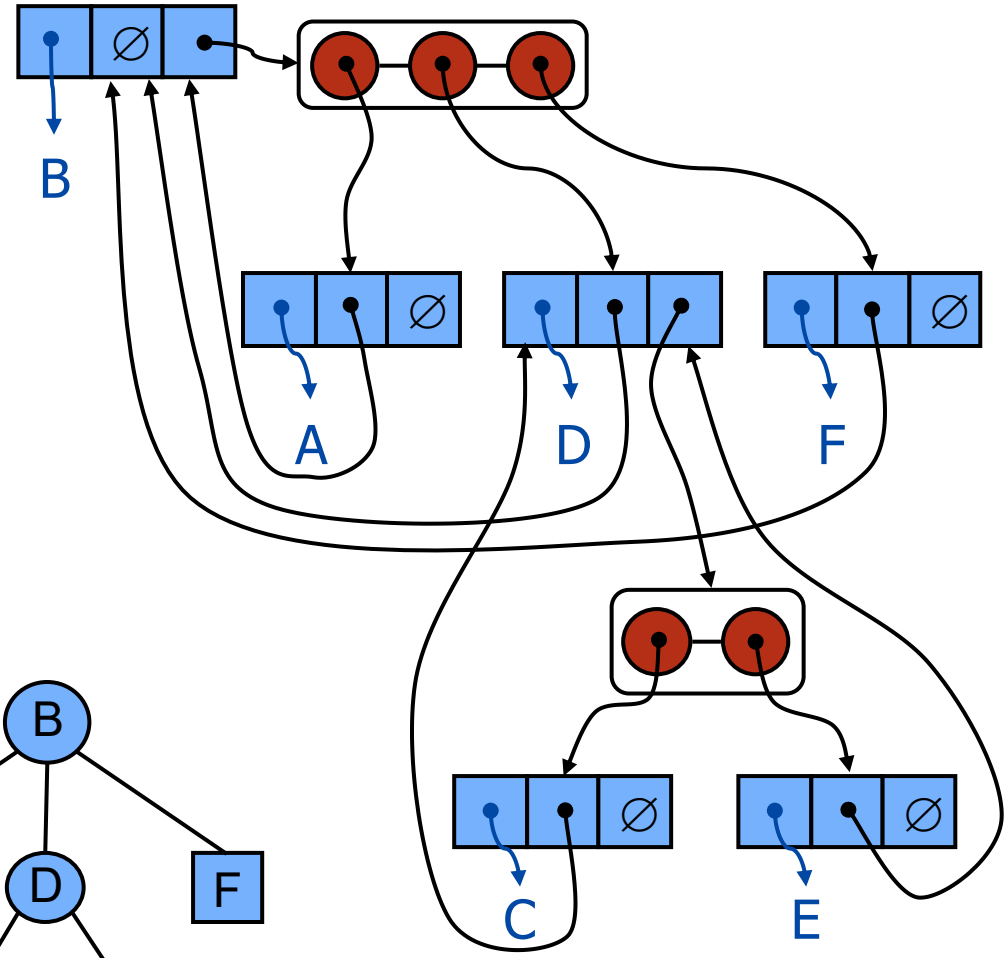
A node is represented by an object storing

- Element
- Parent node
- Sequence of children

Node objects implement the Position ADT

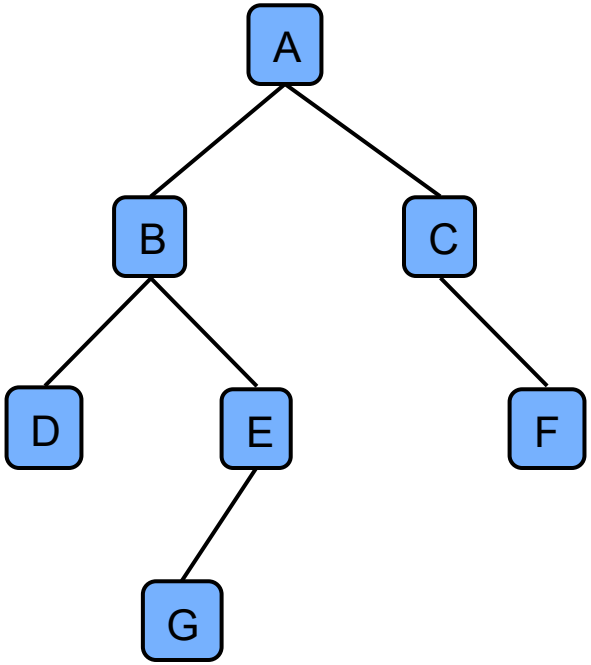






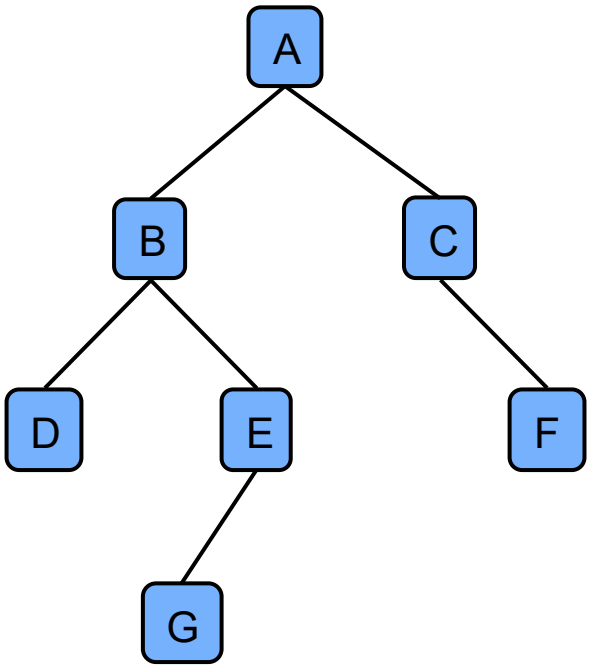
Examples of recursive code on trees

```
def depth(v)
  # compute the depth of v

      if v.parent = null then
        # root's depth is 0
            return 0
      else
            return depth(v.parent) + 1.
```



Examples of recursive code on trees

```
def height(v)
  # compute height of subtree at v

    if v.isExternal() then
      # a leave's height is 0
        return 0
    else
        h ← 0
        for each child w of v
            h ← max(h, height(w))
        return h + 1
```



Complexity analysis of recursive algorithms on trees

Sometimes, the method may call itself on all children

- In worst case, do a call on every node
- If the work done, *excluding the recursion*, is constant per call, then the total cost is linear in the number of nodes

Sometimes, the method calls itself on at most one child

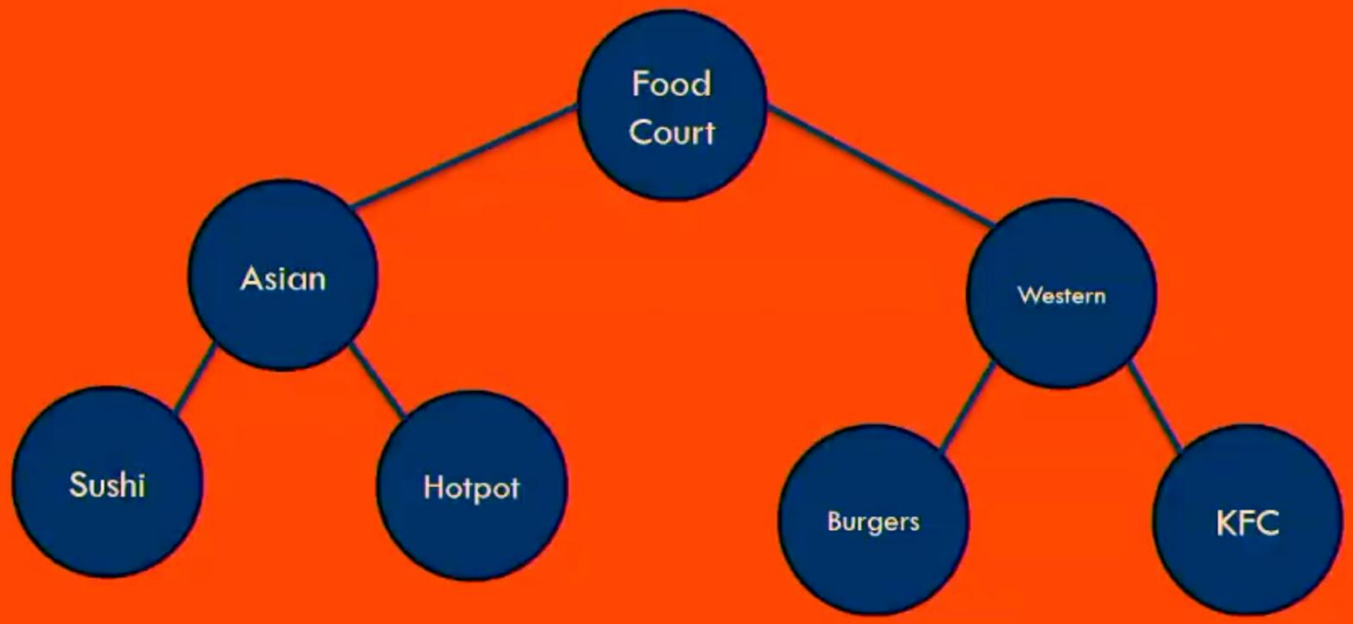
- In worst case, do one call at each level of the tree
- If the work done, *excluding the recursion*, is constant per call, then the total cost is linear in the height of the tree

© Goodrich, Tamassia, Goldwasser

Discovery Activity

You are at Broadway food court and are deciding what to eat. Think of all the options as part of a tree. What would this tree look like and what kind of traversal would you use to choose what to eat?

Discovery Activity



Preorder – If you have an idea of what you want

Inorder – If you want to compare similar categories

Postorder – If you want to check all menu items first

Fully anonymous feedback form for COMP9123 students

Fully anonymous feedback form for COMP9123 students – Fill out form

Updates: